

Lecture 20/Module 2

Transient Analysis of First and Second Order Circuit

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Today's Topics

- General Second Order Circuit

General Second Order circuits

1. We first determine the initial conditions and the final value.
2. We turn off the independent sources and find the form of the transient response by applying KCL and KVL. Once a second-order differential equation is obtained, we determine its characteristic roots. Depending on whether the response is overdamped, critically damped, or underdamped, we obtain with two unknown constants.
3. We obtain the steady-state response.
4. The total response is now found as the sum of the transient response and steady-state response
5. We finally determine the constants associated with the transient response by imposing the initial conditions and determined in step 1.

We can apply this general procedure to find the step response of any second-order circuit, including those with op amps.

Exercise

Find the complete response v and then i for $t > 0$.

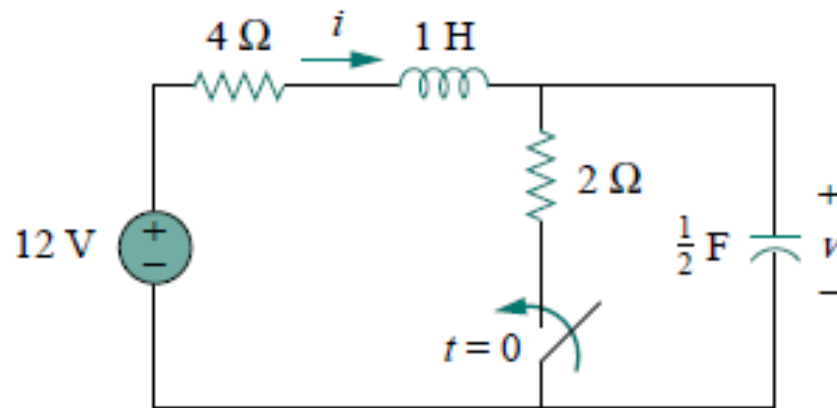


Figure 8.25 For Example 8.9.

Solution:

We first find the initial and final values. At $t = 0^-$, the circuit is at steady state. The switch is open, the equivalent circuit is shown in Fig. 8.26(a). It is evident from the figure that

$$v(0^-) = 12 \text{ V}, \quad i(0^-) = 0$$

At $t = 0^+$, the switch is closed; the equivalent circuit is in Fig. 8.26(b). By the continuity of capacitor voltage and inductor current, we know that

$$v(0^+) = v(0^-) = 12 \text{ V}, \quad i(0^+) = i(0^-) = 0 \quad (8.9.1)$$

To get $dv(0^+)/dt$, we use $C dv/dt = i_C$ or $dv/dt = i_C/C$. Applying KCL at node a in Fig. 8.26(b),

$$i(0^+) = i_C(0^+) + \frac{v(0^+)}{2}$$

$$0 = i_C(0^+) + \frac{12}{2} \quad \Rightarrow \quad i_C(0^+) = -6 \text{ A}$$

Hence,

$$\frac{dv(0^+)}{dt} = \frac{-6}{0.5} = -12 \text{ V/s} \quad (8.9.2)$$

The final values are obtained when the inductor is replaced by a short circuit and the capacitor by an open circuit in Fig. 8.26(b), giving

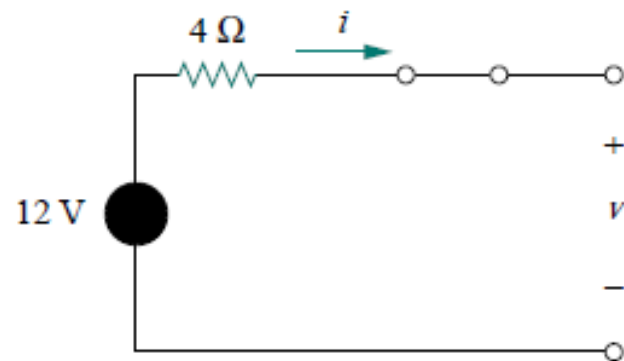
$$i(\infty) = \frac{12}{4+2} = 2 \text{ A}, \quad v(\infty) = 2i(\infty) = 4 \text{ V} \quad (8.9.3)$$

Next, we obtain the natural response for $t > 0$. By turning off the 12-V voltage source, we have the circuit in Fig. 8.27. Applying KCL at node a in Fig. 8.27 gives

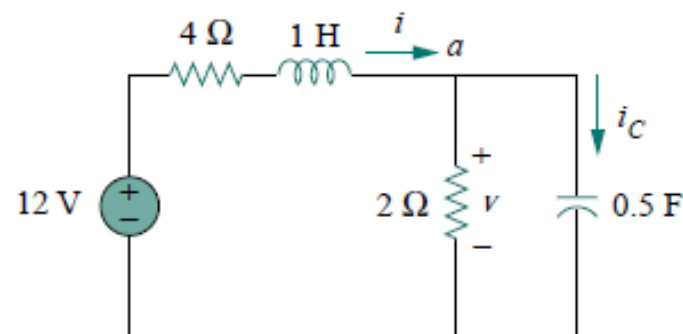
$$i = \frac{v}{2} + \frac{1}{2} \frac{dv}{dt} \quad (8.9.4)$$

Applying KVL to the left mesh results in

$$4i + 1 \frac{di}{dt} + v = 0 \quad (8.9.5)$$



(a)



(b)

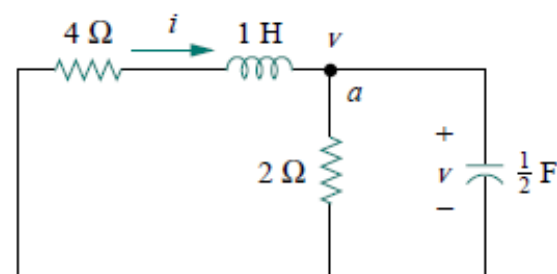


Figure 8.27

Obtaining the natural response for Example 8.9.

Since we are interested in v for the moment, we substitute i from Eq. (8.9.4) into Eq. (8.9.5). We obtain

$$2v + 2\frac{dv}{dt} + \frac{1}{2}\frac{dv}{dt} + \frac{1}{2}\frac{d^2v}{dt^2} + v = 0$$

or

$$\frac{d^2v}{dt^2} + 5\frac{dv}{dt} + 6v = 0$$

From this, we obtain the characteristic equation as

$$s^2 + 5s + 6 = 0$$

with roots $s = -2$ and $s = -3$. Thus, the natural response is

$$v_n(t) = Ae^{-2t} + Be^{-3t} \quad (8.9.6)$$

where A and B are unknown constants to be determined later. The forced response is

$$v_f(t) = v(\infty) = 4 \quad (8.9.7)$$

The complete response is

$$v(t) = v_n + v_f = 4 + Ae^{-2t} + Be^{-3t} \quad (8.9.8)$$

We now determine A and B using the initial values. From Eq. (8.9.1), $v(0) = 12$. Substituting this into Eq. (8.9.8) at $t = 0$ gives

$$12 = 4 + A + B \quad \Rightarrow \quad A + B = 8 \quad (8.9.9)$$

Taking the derivative of v in Eq. (8.9.8),

$$\frac{dv}{dt} = -2Ae^{-2t} - 3Be^{-3t} \quad (8.9.10)$$

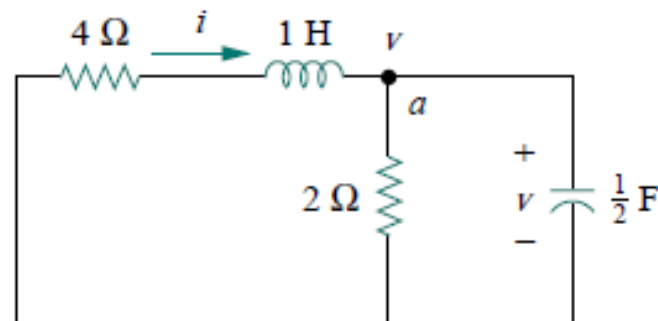


Figure 8.27 Obtaining the natural response for Example 8.9.

Substituting Eq. (8.9.2) into Eq. (8.9.10) at $t = 0$ gives

$$-12 = -2A - 3B \quad \implies \quad 2A + 3B = 12 \quad (8.9.11)$$

From Eqs. (8.9.9) and (8.9.11), we obtain

$$A = 12, \quad B = -4$$

so that Eq. (8.9.8) becomes

$$v(t) = 4 + 12e^{-2t} - 4e^{-3t} \text{ V}, \quad t > 0 \quad (8.9.12)$$

From v , we can obtain other quantities of interest by referring to Fig. 8.26(b). To obtain i , for example,

$$\begin{aligned} i &= \frac{v}{2} + \frac{1}{2} \frac{dv}{dt} = 2 + 6e^{-2t} - 2e^{-3t} - 12e^{-2t} + 6e^{-3t} \\ &= 2 - 6e^{-2t} + 4e^{-3t} \text{ A}, \quad t > 0 \end{aligned} \quad (8.9.13)$$

Notice that $i(0) = 0$, in agreement with Eq. (8.9.1).

Thank You